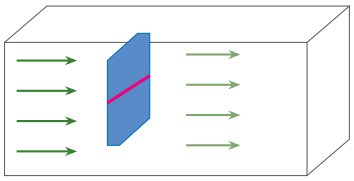
3D FLOW PAST OBJECTS AND THE FLOW TANK

A plate across a stream in a 3D tank (IBFE/10): the gmsh volume mesh, inflow on six sides, the tether band, the initial coordinate mapping in full, the two-material plate, what a 3D run costs, the tether force as drag, and the horseshoe-crab example written up for the first time



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Using IB Software • Part IV • Chapter 22

Claude (Anthropic) was used to organize, combine, and format more than twelve years (2013–2026) of the instructor's tutorial notes from courses and workshops at several universities.

LEARNING GOALS

By the end of this chapter you should be able to:

1. set up a 3D flow tank: a box that is not periodic, a stream prescribed on all six faces that starts from rest, three grid levels, and a plate standing across the stream;
2. build the plate's volume mesh in gmsh (points, lines, a surface, an extrusion, a physical volume), save it in a form libMesh reads, and know what its tetrahedra cost and where its bad elements are;
3. write the tether band and the stress functions of IBFE/10, and use the initial coordinate mapping correctly: one mesh file, placed where you want, with the tether placed with it;
4. give one mesh two materials by the reference position, as the 2017 version did;
5. estimate what a 3D run costs in time and disk, run IBFE/10, read its tether force as minus the drag, and tell the force of an accelerating stream from the force of a steady one;
6. look at a 3D run in VisIt with a slice and a displaced mesh, and describe the horseshoe-crab example, what its code measures, and what must be fixed before it is run again.

> Nothing in the IBFE code changes between 2D and 3D except `NDIM`; what changes is the mesh, the boundary conditions, the cost, and how you look at the result.

WHERE THIS CHAPTER COMES FROM

- "7-IBFE 3D Flow Tank" (Feb 2017, edited 2018; 16 slides; the lab's weekly challenge): goal, domain and plate dimensions, the gmsh GUI route in seven steps with screenshots, the input, "Translate FE boundary" with the coordinate mapping, the tether band, the two-stiffness stress functions. Its example zip `Example_IBFE_FlowTank3d-mid_plate` (2017) is in the 2016-11 set.
- "3-IBFE 3D Flow Past Objects" (2020; 16 slides; `IBFE_Lecture_3`): the same deck without the translation and the two materials ("that feature is not working as it used to in newer versions of IBAMR"), the tank moved to $-8 < x < 24$, the example named `3-IBFE-Example_FlowTank-MidPlate`. Its first git version (2020-05-26) still has the removed slides; both are quoted here.
- The 2018–20 Dogwood deck (private): the "3D IBFE Flow Tank" and "3D IBFE Free Flow" set-ups with their numbers (tank spacing $160/256$ mm, stream 130 mm/s), which are the horseshoe-crab example's input.
- The horseshoe-crab example `3-IBFE-Example-HorseshoeCrab` (2018–2020; course repository): a crab mesh in a free stream at 13 cm/s, with force, velocity and work integrals. No slides were ever written for it; this chapter is the first write-up (slides 30–33).
- Example IBFE/10 (this series): the course program revised, run with IBAMR 0.16.0 and 0.19.0, with the 2017 options restored as input keys and re-tested.
- New here: the runs and their figures; the tether force every 0.05; the cost of the run; the coordinate mapping tested in 0.19.0 (it works); the 2017 input's tank, which was already $-8 < x < 24$ though its slides said $-16 < x < 16$; the gmsh slides' menu paths corrected; the horseshoe-crab code read and its bugs listed.

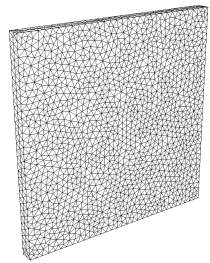
THREE THINGS CHANGE IN 3D

	in 2D (chapters 17–21)	in 3D (this chapter)
cost	a 128×128 fine grid on two levels (IBFE/08), 15 ms per step	$512 \times 256 \times 256$ fine cells if the fine level covered the tank; three levels keep it to the plate and its wake, and the run still takes 0.6 s per step on four cores: 52 minutes for 0.5 time units, 3.5 hours for the course's 2
the mesh	a `.mat` or `.msh` of triangles from MATLAB or gmsh	a volume mesh: a surface extruded into a slab, tetrahedra about DX across, saved as MSH 2.2; some of them flat
boundary conditions	periodic, or a 2D tank (chapter 7)	six faces; the stream on all of them; nothing periodic
looking at it	the whole field in one plot	slices, a displaced mesh, a camera (chapters 27–28)
the code	`NDIM = 2`	`NDIM = 3`: the same functions, `s(2)` and `X(2)` now mean something, `main3d`

The input file knows: `domain_boxes` has three indices, the refinement ratios three entries, the boundary conditions six faces, and a `VelocityBcCoefs_2` block exists.

THE SET-UP OF IBFE/10

tank	$32 \times 16 \times 16$: $-8 < x < 24$, $-8 < y, z < 8$; not periodic
grid	$32 \times 16 \times 16$ cells on level 0 ($\mathrm{DX}_0 = 1$); three levels at ratio 4; $\mathrm{DX} = 1/16 = 0.0625$ on the finest, which covers the plate and the flow near it
fluid	$\rho = 1$, $\mu = 0.01$ (`MU = 1/Re` with `Re = 100`); with the plate's width $D = 2$ and $U = 0.1$, $\rho U D/\mu = 20$
stream	$(U \tanh 2t, 0, 0)$ on all six faces, $U = 0.1$: from rest to 96% of U by $t = 1$, 76% at $t = 0.5$
plate	0.125 thick ($0 < x < 0.125$), 2×2 ($-1 < y, z < 1$); `plate2.msh`, 3386 nodes, 11481 tetrahedra
material	compressible neo-Hookean, $\mathbb{P} = \mu_s \mathbb{F} + (-\mu_s + \beta \ln J)\mathbb{F}^{-T}$, $\mu_s = 0.538$, $\beta = 0.404$ (chapter 19)
tether	the band $\lvert s_2 \rvert < 0.1$ across the middle, $\kappa_s = 10^5$, pulled to its starting position; the two flaps above and below are free
time	`DT = 1e-4`, 5000 steps to $t = 0.5$ (the course ran to 2); 21 frames, 100 force lines



The plate's mesh as libMesh reads it: about two tetrahedra through the thickness.

- "Create a plate in a 3D flow tank. Let the plate be tethered along its centerline. Move the flow from left to right along x-axis" (2020, slide 2).
- "Strategy for domain and plate dimensions: . . . dx = 16/256 = 0.0625. Make a plate that is 0.125 x 2 x 2 units" (slide 3).

THE STREAM ON SIX FACES

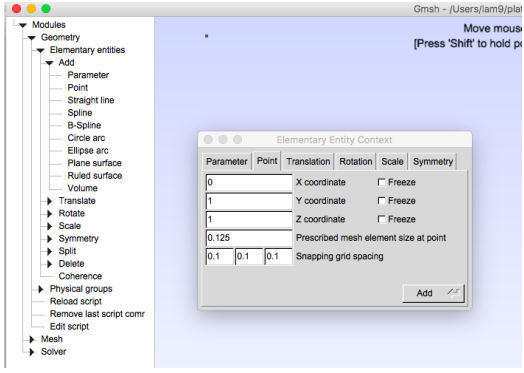
```
U_FLOW = 0.1          // flow speed far from the plate
...
VelocityBcCoefs_0 {   // the x-component of the velocity
   U0 = U_FLOW        // expressions can use only constants
                      // defined in their own block
   acoef_function_0 = "1.0"   ...   acoef_function_5 = "1.0"
   bcoef_function_0 = "0.0"   ...   bcoef_function_5 = "0.0"
   gcoef_function_0 = "U0*tanh(2*t)"
   gcoef_function_1 = "U0*tanh(2*t)"
   gcoef_function_2 = "U0*tanh(2*t)"
   gcoef_function_3 = "U0*tanh(2*t)"
   gcoef_function_4 = "U0*tanh(2*t)"
   gcoef_function_5 = "U0*tanh(2*t)"
}
VelocityBcCoefs_1 { ... gcoef_function_0 = "0.0" ... }   // y
VelocityBcCoefs_2 { ... gcoef_function_0 = "0.0" ... }   // z
...
CartesianGeometry {
   domain_boxes = [ (0,0,0),(2*N - 1,N - 1,N - 1) ]
   x_lo = -0.5*L,-0.5*L,-0.5*L
   x_up =  1.5*L, 0.5*L, 0.5*L
   periodic_dimension = 0,0,0
}
```

- Faces 0–5 are $x_{\mathrm{lo}}, x_{\mathrm{hi}}, y_{\mathrm{lo}}, y_{\mathrm{hi}}, z_{\mathrm{lo}}, z_{\mathrm{hi}}$; each is $a\,u + b\,(\text{normal stress}) = g$, and $a = 1$, $b = 0$ makes g the velocity (chapter 7). The whole tank, walls included, moves with the stream: a towing tank seen from the plate.
- The same value on the inflow face and the outflow face means the outflow is *prescribed*, not free: the wake must get back to U within the 24 units downstream. At $\mathrm{Re} = 20$ it does; at a higher Reynolds number use a traction outflow ($b = 1$) as chapter 7 did.
- $\tanh 2t$ starts the stream smoothly; an impulsive start would put a large acceleration reaction on the plate in the first step (slide 22).
- The course wrote `0.1*tanh(2*t)` into each of the six strings; `U0` copied into the block is the maintainable way, because muParser sees only that block's constants.
- The plate stands 8 units from the inflow face and 24 from the outflow. The 2017 version put it 1 unit from the inflow (slide 16).

MAKING THE PLATE IN GMSH (1): A SQUARE

"Step 1: Make the mesh in gmsh. The resulting file should be similar to plate2.msh in the Example folder. Remember that we want the element size to be about dx = 0.0625. We can make this by making a square and then extruding it into the third dimension" (2020, slide 4).

1. **File** → **New**; **Geometry** → **Elementary entities** → **Add** → **Point**: the four corners $(0, 1, 1)$, $(0, -1, 1)$, $(0, -1, -1)$, $(0, 1, -1)$, each with "Prescribed mesh element size at point" 0.0625. Click **Add** for each.
2. **Add** → **Straight line**: click one point, then the next; four times round. "Left click and drag to rotate the image in 3D."
3. **Add** → **Plane surface**: click a line, the whole loop lights up; press **e** to end the selection. "A grid should appear over the plane" (the dashed cross marks the surface).

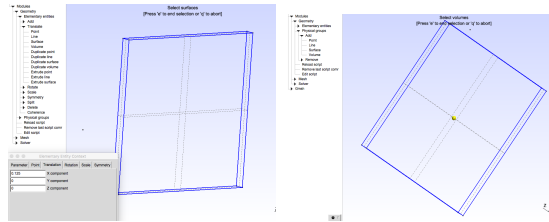


The point dialog of the 2020 deck. Its "Prescribed mesh element size" reads 0.125 in the screenshot while the text says 0.0625; `plate2.msh` has elements about 0.07 across, the text's value.

Screenshot from the 2020 deck (Laura Miller, gmsh on macOS).

MAKING THE PLATE IN GMSH (2): EXTRUDE, VOLUME, PHYSICAL GROUP

4. **Elementary entities** → **Translate** → **Extrude surface**: 0.125 in x; click the surface (it turns red), press **e**. “This will make plate 2 elements thick.” (The 2020 deck wrote **Add** → **Translate**; Extrude lives under Translate, not Add.)
5. **Add** → **Volume**: click the dashed boundary of the slab, press **e**. (The extrusion already makes the volume; this step is harmless and redundant.)
6. **Physical groups** → **Add** → **Volume**: click the yellow dot in the slab, press **e**; name it `plate2`. The name becomes the subdomain’s name in libMesh (chapter 18).



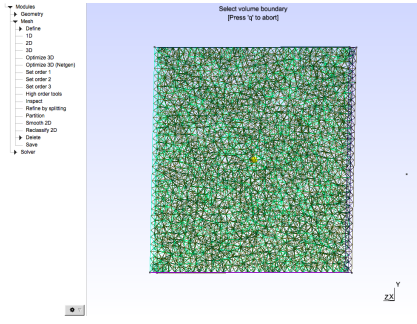
The extrude dialog (0.125 along x); the physical-volume pick (the yellow dot).

Screenshots from the 2020 deck.

MAKING THE PLATE IN GMSH (3): MESH AND SAVE

7. **Mesh** → **3D**, then **File** → **Save mesh**. “This leaves only the 3D mesh and no 1D or 2D meshes that libmesh will reject”: with physical groups defined, gmsh saves only the elements that belong to one, so the lines and the surfaces stay out of the file.

- The deck’s “saves the mesh of the physical group as group_name.msh” is not right: the file takes the *project’s* name; you choose it in **File** → **Save as**. The physical group’s name goes inside the file, as the subdomain’s name.
- gmsh 4 saves MSH 4.1 by default; libMesh 1.7 reads MSH 2.2 and 4 as text, not binary. In **Save as** choose “Version 2 ASCII”, or set `Mesh.MshFileVersion = 2.2` in a `.geo` file (chapter 18).
- The same seven steps as a script: a `.geo` with four `Point`s, four `Line`s, a `Line Loop`, a `Plane Surface`, `Extrude {0.125, 0, 0} { Surface{1}; }` and `Physical Volume("plate2")`, then `gmsh -3 -format msh22 plate.geo`: chapter 18’s route, which can be repeated and versioned.



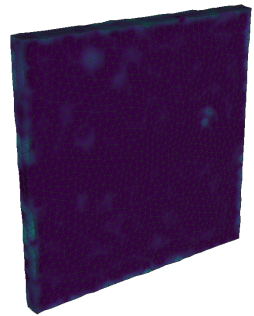
The 3D mesh in gmsh (2020 deck): tetrahedra on the slab.

THE MESH AS LIBMESH READS IT

- `plate2.msh`: MSH 2.2 text, 3386 nodes, 11481 elements of gmsh type 4 (four-node tetrahedra, `TET4`), one physical volume `plate2`. Volume 0.5: a tetrahedron of edge h has volume $h^3/(6\sqrt{2})$, so the mean edge is about 0.07, the grid spacing, and there are two elements through the 0.125 thickness.
- `MFAC` in the input ("ratio of Lagrangian mesh width to Cartesian mesh width") does nothing here: it sizes meshes built in the program (IBAMR's `ex2`); a mesh read from a file is as fine as gmsh made it (chapter 18).
- Reading it: `mesh.read("plate2.msh")`, `mesh.prepare_for_use()`; the name comes from `MESH_FILENAME` in the input. The 2017 deck noted "the mesh name also appears in the main.C file, but I think those lines are obsolete"; the 2020 deck says it is "given in both": it is the input key's default value.
- With `IB_point_density = 1` and `QGRID`, each tetrahedron gets interaction points about DX apart (slide 19): about one per element here, which is what a body resolved by the grid needs; a body finer than the grid needs no more (chapter 17).

THE MESH (2): THE FLAT ELEMENTS

- 80 of the 11481 tetrahedra have a shape quality below 0.05 (a regular tetrahedron is 1) and 4 are below 0.01: slivers that the extrusion's two layers of tetrahedra produce near the plate's faces.
- They are where $J = \det \mathbb{F}$ strays from 1 in the run (0.21 to 2.1 at $t = 0.5$ while the rest of the plate stays near 1), and where a few nodes move 0.05 when the plate's edge moves 0.01 (slide 23). A flat element has almost no volume, so a small displacement of one node changes its J a lot; the stress there is large but acts on almost nothing, and the run survives. With a larger time step or a stiffer material, J in a sliver goes negative first, $\ln J$ is undefined, and the run ends.
- **Mesh** $\rightarrow$ **Optimize 3D** in gmsh improves them; three layers instead of two avoid them; `Mesh.OptimizeNetgen = 1` in a `.geo` does it in a script. Chapter 16's checks (a short run, the range of J) catch them before a long run.



The plate at $t = 0.5$ coloured by its downstream displacement on a scale to 0.05: almost everything is dark (below 0.012); the brighter spots are the sliver tetrahedra's nodes.

THE CODE (1): THE TETHER BAND AND THE STRESS FUNCTIONS

```
static double kappa_s = 1.0e5;  static double tether_half_width = 0.1;  static double translate_x = 0.0;   // from input3d
// Target ("tether") force on the plate, per unit reference volume. s is the reference position of a point of the
// plate (its position in the mesh file); X is where that point is now. Only the band |s(2)| < tether_half_width is tethered.
void target_force_function(VectorValue<double>& F, const TensorValue<double>& /*FF*/, const libMesh::Point& X,
                           const libMesh::Point& s, Elem* const /*elem*/, const std::vector<const std::vector<double>*>& /*var_data*/
  ,
                           const std::vector<const std::vector<VectorValue<double> >*>& /*grad_var_data*/, double /*time*/, void* /*
  ctx*/)
{
    double kappa = 0.0;                                    // no tether force outside the band
    if (std::abs(s(2)) < tether_half_width) kappa = kappa_s;
    libMesh::Point s_target = s;                           // the tether location...
    s_target(0) += translate_x;                            // ...after the translation (none by default)
    F = kappa * (s_target - X);
}
void PK1_dev_stress_function(TensorValue<double>& PP, const TensorValue<double>& FF, ..., const libMesh::Point& s, ...)
{
    const bool leaf = two_materials && std::abs(s(2)) >= stem_half_width;
    const double mu = leaf ? mu_leaf : mu_stem;
    PP = mu * FF;
}
void PK1_dil_stress_function(TensorValue<double>& PP, const TensorValue<double>& FF, ..., const libMesh::Point& s, ...)
{
    const bool leaf = two_materials && std::abs(s(2)) >= stem_half_width;
    const double mu = leaf ? mu_leaf : mu_stem;  const double beta = leaf ? beta_leaf : beta_stem;
    const double J = FF.det();  J_minimum = std::min(J_minimum, J);  J_maximum = std::max(J_maximum, J);   // for output_data
    PP = (-1.0 * mu + beta * std::log(J)) * tensor_inverse_transpose(FF, NDIM);
}
```

THE CODE (1B): THE TETHER IN ITS THREE VERSIONS

2017 (“main.C – Target forces”; the old `system_data` signature):

```
libMesh::Point s_dump;
double kappa;
s_dump = s; // this is the tether location before translation
s_dump(0) = s(0) - 7; // now the desired location is translated.
// apply tether force only if z-coordinate of original position
   is within 0.1 of center.
if (abs(s(2)) < .1) { kappa = kappa_s; }
else { // no tether force, set tether stiffness to zero.
       kappa = 0; }
F = kappa * (s_dump - X); // Note x is actual position, s_dump is
    desired position
```

2020 (slide 14; the translation gone, `s_dump` left behind):

```
libMesh::Point s_dump;
double kappa;
s_dump = s;
if (abs(s(2)) < .1) { kappa = kappa_s; }
else { kappa = 0; }
F = kappa * (s - X); // Note x is actual position, s is desired
   position
```

- The stress functions are chapter 19’s compressible neo-Hookean pair, with the material chosen by where the point is; the tether is chapter 19’s body force restricted to a band of the reference configuration.
- “Apply target force at the midline of plate using target_force_function. Applied in band running from $-.1$ to $.1$ in z-direction” (2020, slide 13).
- The 2017 version is the right one when the plate is moved: the target is `s` *shifted*, as the mapping shifted the mesh (slide 15). The 2020 version is right only because the mapping was removed with it.
- IBFE/10 keeps one function for both cases: `s_target(0) += translate_x`, with `translate_x = 0` by default; the band width is an input key; `std::abs` replaces `abs`, which on some compilers truncates a double to an integer and makes every point “inside”.
- The comment “Note x is actual position” names the variable `X`; the lower-case `x` in the 2017 and 2020 slides is a slip that survived three years.

THE CODE (2): THE INITIAL COORDINATE MAPPING

IBFE/10:

```
// Initial coordinate mapping: where each point of the mesh
  starts.
// s is the position in the mesh file and X is the starting
  position in the fluid.
void coordinate_mapping_function(libMesh::Point& X, const libMesh
  ::Point& s, void* /*ctx*/)
{
    X(0) = s(0) + translate_x;
    X(1) = s(1);
    X(2) = s(2);
}
...
if (ModelData::translate_x != 0.0)
{
    ib_method_ops->registerInitialCoordinateMappingFunction(
  coordinate_mapping_function);
}
ib_method_ops->initializeFEEquationSystems();
```

The 2017 deck (slide 13, “main.C – Translate FE boundary”): “We made our FE mesh at the origin. We want to shove the plate left so it’s at the beginning of the flow tank. The coordinate_mapping_function does this”:

```
void coordinate_mapping_function(libMesh::Point& X, const
  libMesh::Point& s, void* /*ctx*/)
{
    X(0) = s(0) - 7; // translated 7 units left along x-axis
    return;
}
...
ib_method_ops->registerInitialCoordinateMappingFunction(
  coordinate_mapping_function);
```

- The signature: the starting position `X` to fill, the file’s `s`, and `ctx`. Chapter 21’s two-plate example used it to turn a plate upright and place two copies; chapter 17 introduced it with the rotated plate.

THE COORDINATE MAPPING (2): WHAT IT DOES, AND WHAT MUST FOLLOW IT

- What it does. Before the first step, IBFE sets every node's position $\mathbf{X}$ from its file coordinates $\mathbf{s}$ through this function (the default is $\mathbf{X} = \mathbf{s}$). Only $\mathbf{X}$ moves afterwards; $\mathbf{s}$ stays the file's coordinates for the whole run and in every callback: the stress functions, the forces, the post-processing.
- So the tether must be mapped too. A target of $\mathbf{s}$ pulls the plate back to the file's position. The 2017 code wrote `s_dump(0) = s(0) - 7` in the tether; IBFE/10 adds `translate_x` to `s_target`. Any function that compares $\mathbf{X}$ to $\mathbf{s}$ (a tether, a prescribed motion, chapter 21's two copies) needs the same mapping; a material chosen by $\mathbf{s}$ does not, because the file's coordinates still say where in the body a point is.
- Set every component. The 2017 function set `X(0)` only; `X` arrives holding $\mathbf{s}$, so that works, but write all three (IBFE/10) and nobody has to know.
- `dX` in the output is $\mathbf{X} - \mathbf{s}$: it includes the shift. In VisIt the displaced mesh is in the right place; a plot of `dX_0` on a plate moved by 4 reads about 4 everywhere, and the bend is the small variation on top.
- Register before `initializeFEEquationSystems()`, once per part (`..., part`) for several); a mapping registered later is ignored.
- A rotation is the same mechanism: $\mathbf{X} = R\,\mathbf{s}$ with a rotation matrix R, and the tether target $R\,\mathbf{s}$. The horseshoe crab's "0 degrees" (slide 30) is where that goes.

THE COORDINATE MAPPING (3): “NOT WORKING AS IT USED TO”, RE-TESTED

- The 2020 deck dropped the translation: “Previously, we translated the plate to the left. That feature is not working as it used to in newer versions of IBAMR” (slide 3), moved the tank from $-16 < x < 16$ to $-8 < x < 24$ instead, and commented the registration out (`//ib_method_ops->registerInitialCoordinateMappingFunction(...)` in the course’s `main.C`).
- What the 2017 input actually said. Its slide 3 reads “The domain goes from $-16 < x < 16$ . . . translate 7 units in the negative x-direction”. But the `input3d` in the 2017 zip already has `x_lo = -0.5*L`, `x_up = 1.5*L`: the tank was $-8 < x < 24$ then too, and the plate shifted by -7 stood *one unit* from the inflow face, inside the kernel’s reach of the boundary where the velocity is prescribed. A plate that cannot move its fluid is a plausible “not working”.
- Re-tested with IBAMR 0.19.0. `TRANSLATE_X = 4.0` (the plate 12 units from the inflow face), run to $t = 0.1$: the plate sits at $x = 4$ (its mean x in `output.ex2` is 4.06), and the tether force at $t = 0.1$ is the same as in the default run to four digits (-0.1064 against -0.1064). The mapping works; in a tank this large the plate’s position hardly matters.
- When to use it. One mesh file, many placements: an angle of attack, a position in the tank, chapter 21’s second copy. The alternative, editing the mesh in gmsh, changes a file that others share.

> Rule: map `s` wherever the code compares positions, and keep the body away from a face where the velocity is prescribed: at least a few kernel widths, and much more if its wake must leave through the outflow.

THE 2017 TWO-MATERIAL PLATE

"Let the plate have 2 stiffnesses (middle half is different than ends)" (2017, slide 2); in the code the band is $|s_2| < 0.25$, the middle *quarter*:

```
// 2017 main.C
if (abs(s(2)) < .25) { mu = mu_stem; beta = beta_stem; }
else                 { mu = mu_leaf; beta = beta_leaf; }
// 2017 input3d
NU=0.3
E_STEM=7000
E_LEAF=140000
MU_LEAF = E_STEM/(2.0*(1.0+NU))      // 1e2
MU_STEM = E_LEAF/(2.0*(1.0+NU))      // 1e5
BETA_STEM = (E_STEM*NU) / ((1.0 + NU)*(1.0 - 2.0*NU))
BETA_LEAF = E_LEAF*NU/((1.0 + NU)*(1.0 - 2.0*NU))
KAPPA_S = 1e6
```

- $\mu = E/(2(1+\nu))$ and $\beta = E\nu/((1+\nu)(1-2\nu))$ are the shear modulus and the first Lamé parameter of a linear solid with Young's modulus E and Poisson's ratio ν: the way to choose a neo-Hookean material's two constants from numbers you can look up.

- The swap. Since 2017 the two `MU` lines use each other's E (`MU_STEM` from `E_LEAF`) while the `BETA` lines use their own, and the comments (10^2, 10^5) match neither. IBFE/10 keeps the swap so its runs match the course's: the plate has $\mu_s = 0.538$ ($E = 1.4$) and $\beta = 0.404$ ($E = 0.7$), which together correspond to $E \approx 1.31$, $\nu \approx 0.21$. Fix it for your own runs.
- Restored as an input option. `TWO_MATERIALS = TRUE` with `STEM_HALF_WIDTH = 0.25`: stem $\mu = 0.538$, $\beta = 0.404$; leaf $\mu = 0.269$, $\beta = 0.808$ (the softer flaps). Run to $t = 0.1$ with 0.19.0: the force on the plate is 9% smaller (-0.097 against -0.106), the flap edges have moved the same 0.001, and J ranges further (0.64 to 2.96).
- Choosing the material by `s` is chapter 19's "a stress that depends on where you are"; a subdomain id from gmsh's physical groups (chapter 18) does the same without a coordinate test.

THE INPUT FILE (1): GRID, TIME AND OUTPUT

```
L = 16.0                    // width of the domain
MAX_LEVELS = 3
REF_RATIO  = 4
N = 16                      // cells across the width on level 0
NFINEST = (REF_RATIO^(MAX_LEVELS - 1))*N    // 256
DX  = L/NFINEST             // 0.0625
MFAC = 3.0                  // no effect: the mesh is a file
PK1_DEV_QUAD_ORDER = "SEVENTH"
PK1_DIL_QUAD_ORDER = "FIFTH"
KAPPA_S = 1e5
TETHER_HALF_WIDTH = 0.1
TRANSLATE_X = 0.0
TWO_MATERIALS = FALSE
STEM_HALF_WIDTH = 0.25
...
DT                          = 1e-4
CFL_MAX                     = 0.2
END_TIME                    = 0.5    // the course: 2.0
VIZ_INTERVAL     = int(END_TIME/(20*DT))   // 21 frames
DATA_INTERVAL    = int(END_TIME/(100*DT))  // 100 lines
RESTART_INTERVAL = int(0.5/DT + 0.5)
GriddingAlgorithm {
   ratio_to_coarser {
      level_1 = REF_RATIO,REF_RATIO,REF_RATIO ... }
   largest_patch_size { level_0 = 256,256,256 }
   smallest_patch_size { level_0 = 8,8,8 } ... }
```

- Three refinement ratios per level, three sizes per patch; `largest_patch_size` 256^3 lets one patch hold a whole level.
- `vorticity_abs_thresh = 0.125,0.25,.5,1,2` tags cells for refinement where $|\boldsymbol{\omega}|$ exceeds the level's threshold; with $\mathrm{Re} = 20$ the vorticity is weak and the plate's own cells (tagged because they hold Lagrangian points) decide most of the fine level (slide 26).
- `PK1_*_QUAD_ORDER` are the stress quadrature orders of chapter 19 (`SEVENTH` and `FIFTH` here, `THIRD` by default); for linear tetrahedra and a smooth stress they are generous.
- $\mathrm{DT} = 10^{-4}$ with $\mathrm{DX} = 1/16$ and $U = 0.1$ is a CFL of 0.00016: the limit is the tether, $\kappa_s = 10^5$, not the flow (chapter 19, slide 26).
- The intervals are written in terms of `END_TIME` and `DT`, so a longer run keeps 20 frames and 100 force lines; the course's `int(0.1/DT)` gave 20 frames for its 2 time units.

THE INPUT FILE (2): THE IBFE INTERACTION KEYS

```
IB_DELTA_FUNCTION           = "IB_4"
SPLIT_FORCES                = FALSE
USE_JUMP_CONDITIONS         = FALSE
USE_CONSISTENT_MASS_MATRIX = TRUE
IB_POINT_DENSITY            = 1.0
...
IBFEMethod {
   IB_delta_fcn                 = IB_DELTA_FUNCTION
   split_forces                 = SPLIT_FORCES
   use_jump_conditions          = USE_JUMP_CONDITIONS
   use_consistent_mass_matrix =
  USE_CONSISTENT_MASS_MATRIX
   IB_point_density             = IB_POINT_DENSITY
   IB_quad_type                 = "QGRID"   // or "QGAUSS
  "
}
```

key (read by `getFromInput`)	what it sets
`IB_delta_fcn`	the kernel for interpolation and spreading (`IB_4` here; chapter 17)
`IB_point_density`	interaction points per DX along each direction of an element: 1 gives about one point per DX, 2 twice as many
`IB_quad_type`, `IB_quad_order`	the rule that places them: `QGRID` on a grid inside the element (adaptive to the density), `QGAUSS` at Gauss points of the given order
`IB_use_nodal_quadrature`	the nodes themselves as interaction points, with a lumped mass matrix (`QTRAP`, first order)
`interp_*`, `spread_*`	the same keys for interpolation and spreading separately; `IB_*` sets both
`split_forces`, `use_jump_conditions`	the boundary-term treatment of chapter 19, slide 7
`use_consistent_mass_matrix`	the L^2 projection of the velocity onto the mesh: consistent (a solve) or lumped

RUNNING IT

```
# laptop (autoibamr); on Puma add: module load
  gnu13 openmpi5
source ~/autoibamr/configuration/enable.sh
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j 4
mpirun -np 4 ./build/main3d input3d
# continue a finished run from its restart files:
mpirun -np 4 ./build/main3d input3d restart_IB3d
  5000
```

- On Puma the modules load only on compute nodes: build and run inside an interactive session or the batch job (`job.slurm`: 4 cores, the `standard` partition); chapter 31.
- The restart line continues from step 5000 with the finite element data of the plate reloaded and `output.ex2` appended (chapter 21's rule).

5000 steps, 4 cores (laptop, WSL)	3096 s: 0.62 s per step
the course's 2 time units	20000 steps: about 3.5 hours
`viz_IB3d/` (21 frames of 4.4 MB, plus `output.ex2`, 7.2 MB)	about 100 MB
restart files at the end	10 MB; the folder 115 MB
0.16.0 and 0.19.0	the same tether force to all six digits at $t = 0.5$

WHAT A 3D RUN COSTS

- Why 0.6 s per step: three levels of a $32 \times 16 \times 16$ base, 11481 elements with a point density of 1, and a staggered-grid solve of the whole tank every step. IBFE/08's 2D plate took 15 ms per step. The cell count grows as DX^{-3}, and the work as DX^{-4} once the time step shrinks with the cells.
- Before a long run: a short one (`END_TIME = 0.002`, 20 steps, the "smoke test") to see that the files appear and the force line prints; then a run to 0.1 to time it; then the real one, in a batch job, with restart files written on the way (`RESTART_INTERVAL`).
- The course's `data_dump_interval = 1` called an empty `output_data` every step and printed "Max Jacobian 1" that never changed; IBFE/10 computes the tether force 100 times per run and prints the real range of J.
- Output per frame is the whole hierarchy: keep `VIZ_INTERVAL` coarse and use restarts for the rest. Forty frames of this run are 180 MB; the course's twenty frames of a tank at $\mathrm{DX} = 1/32$ would be eight times that.
- The horseshoe crab (slide 30) is the other end: 50000 steps on 64 cores for up to 72 hours.

WHAT YOU SHOULD SEE (1): THE TETHER FORCE

Every 50 steps the screen shows

```
tether force = -0.106699 0.00112701 0.00150725 J from
                    0.822529 to 1.72649
```

the total tether force on the plate and the range of J since the last line; the force goes to `hier_data_IB3d/tether_force.txt`.

t	$U(t)$	F_x	J range
0.05	0.010	-0.107	0.82–1.73
0.10	0.020	-0.106	0.66–2.51
0.15	0.029	-0.150	
0.20	0.038	-0.192	0.35–2.72
0.25	0.046	-0.194	
0.30	0.054	-0.215	0.52–2.14
0.35	0.060	-0.221	
0.40	0.066	-0.239	
0.45	0.072	-0.263	
0.50	0.076	-0.276	0.21–2.10

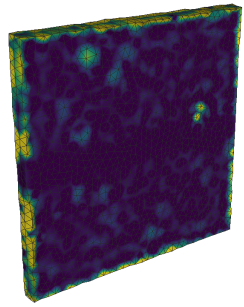
- The tether is what holds the plate against the stream, so minus the tether force is the force of the fluid on the plate: F_x is minus the drag (once the flaps have stopped bending, on the whole plate; before that, on the band plus what the flaps hand to it).
- An accelerating stream pushes before it flows. At $t = 0.05$ the stream is at a tenth of its speed and the force is already 0.107, 40% of its value at $t = 0.5$: the fluid around the plate is accelerating ($\dot{U} = 2U\,\mathrm{sech}^2 2t$, 0.2 at the start) and the plate must push it. That force falls as $\dot{U}$ falls (to 0.084 at $t = 0.5$) while the steady drag rises with U; the sum is what the table shows. A run that starts the stream impulsively sees this force in one step, with the tether stiffness deciding how large.
- F_y and F_z are 0.036 and -0.015 at $t = 0.5$: the plate is not quite symmetric (the mesh is not), and the top flap bends a little more than the bottom one.
- 0.16.0 and 0.19.0 give the same ten lines to four digits.

WHAT YOU SHOULD SEE (2): THE FLAPS

From `output.ex2` (the free edges are the nodes at $z = \pm 1$ in the mesh file):

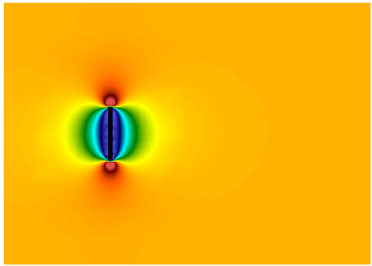
t	top edge Δx	bottom edge Δx	band
0.1	0.0011	0.0010	$< 3 \times 10^{-5}$
0.2	0.0040	0.0035	
0.3	0.0076	0.0065	
0.4	0.0102	0.0085	
0.5	0.0116	0.0094	$< 3 \times 10^{-5}$

- The flaps bend downstream slowly: 0.012 at the edge of a flap 1 long is a small bend, as $\mu_s = 0.54$ against a dynamic pressure of $\rho U^2/2 = 0.003$ predicts. The top flap bends a little more. The tethered band does not move.
- A few nodes move up to 0.05: the sliver tetrahedra of slide 11, which are also where J strays. They are a mesh defect, not a flow feature.
- Softer flaps (`E_STEM = 0.07`) or a faster stream bend more; the course's run to $t = 2$ reaches the steady stream and a steady bend.

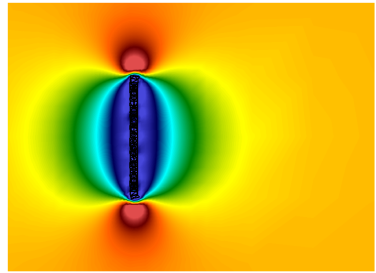


The plate at $t = 0.5$, coloured by its downstream displacement `dX_0` (0 to 0.012), at true displacement, seen from downstream: the free edges have moved most.

WHAT YOU SHOULD SEE (3): THE FLOW



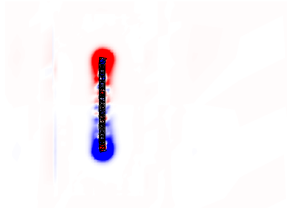
u_x (0 to 0.12) on the slice $y = 0$ at $t = 0.5$; $-4 < x < 10$. The plate's cross-section in black.



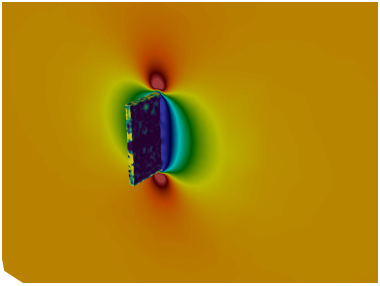
The same, $-2 < x < 4$: the stream goes round the plate's edges, a slow wake behind it.

- At $\mathrm{Re} = 20$ there is no shedding: a steady recirculation forms behind the plate and the wake recovers within a few plate widths. The slice is the x–z plane through the tethered band's middle, so it cuts the band and both flaps.
- Ahead of the plate the fluid slows (stagnation); above and below its edges it speeds up to 0.12, above the stream's 0.076.

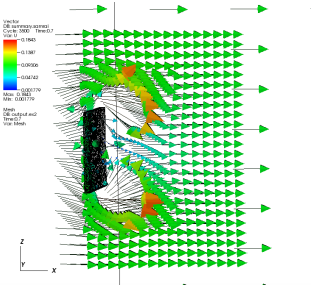
WHAT YOU SHOULD SEE (4): VORTICITY, THE 3D VIEW, AND THE COURSE'S OWN PICTURE



ω_y (± 0.3) on the slice $y = 0$: vorticity of opposite sign shed from the plate's two edges.



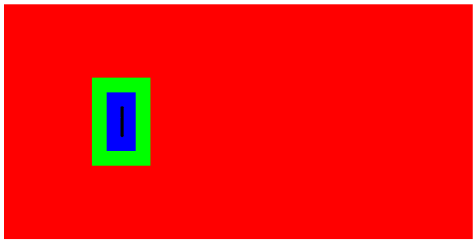
The slice in 3D with the displaced plate coloured by `dX_0`.



The 2020 deck's render (slide 2): velocity vectors on a slice at $t = 0.7$ of the course's run, the plate's mesh in black.

- In 3D the vorticity is a vector; the 2D decks' `Omega` is its component normal to the plane you look at, `Omega[1]` on this slice, and `magnitude(Omega)` is the scalar for a 3D view.
- The course's picture shows the same flow 0.2 later: the vectors bend round the plate and the recirculation behind it shows in the vectors that point upstream.

THE GRID: THREE LEVELS



The tank on the slice $y = 0$ at $t = 0.5$: level 0 (red, $\mathrm{DX} = 1$), level 1 (green, $1/4$), level 2 (blue, $1/16$). The plate's section is the black line inside the blue box.

- The finest level is a box round the plate and its near wake; the rest of the tank is 1 and $1/4$ coarse. With $\mathrm{Re} = 20$ the vorticity thresholds tag little; the Lagrangian points tag the plate's cells, and `tag_buffer = 1` pads them.
- If the whole tank were at $\mathrm{DX} = 1/16$ it would be $512 \times 256 \times 256 = 33$ million cells; the three levels keep it to a few hundred thousand.

THE GRID (2): WHAT TO ADJUST

- A wake that leaves the blue box is computed on the coarser level, with the $1/4$ spacing: fine at this Reynolds number, not at a higher one. The thresholds (`vorticity_abs_thresh`, one per level) and so the box the fine level covers are what you adjust; chapter 16's convergence study says when it is enough.
- `regrid_cfl_interval = 0.5`: the hierarchy is rebuilt whenever a point could have moved half a cell; for a plate that barely moves that happens a few times per run (three regrids in this run's log). A body that swims regrids often, and each regrid costs a few steps' worth of time.
- `tag_buffer = 1` pads every tagged region by one cell of the coarser level; a larger buffer keeps a moving body inside its fine box longer between regrids.
- The ratio 4 between levels is IBAMR's usual choice: 16 cells of the fine level for one coarse cell in 2D, 64 in 3D. A ratio of 2 with more levels refines more gradually at the price of more levels to manage.
- A finer base grid ($N = 32$) with the same three levels doubles the resolution everywhere and costs eight times the cells and sixteen times the work; a fourth level costs only where it is placed.

THE POST-PROCESSING (1): THE TETHER FORCE AS AN INTEGRAL OVER THE PLATE

```
void output_data(..., Mesh& mesh, EquationSystems* equation_systems, const std::string& coords_system_name,
                 const int iteration_num, const double loop_time, const string& data_dump_dirname)
{
    System& X_system = equation_systems->get_system(coords_system_name);          // the current positions X
    NumericVector<double>* X_ghost_vec = X_system.current_local_solution.get();  copy_and_synch(*X_system.solution, *X_ghost_vec);
    const DofMap& dof_map = X_system.get_dof_map();  std::vector<std::vector<unsigned int> > dof_indices(NDIM);
    std::unique_ptr<FEBase> fe(FEBase::build(dim, dof_map.variable_type(0)));
    std::unique_ptr<QBase> qrule = QBase::build(QGAUSS, dim, FIFTH);  fe->attach_quadrature_rule(qrule.get());
    const std::vector<double>& JxW = fe->get_JxW();  const std::vector<libMesh::Point>& q_point = fe->get_xyz();
    const std::vector<std::vector<double> >& phi = fe->get_phi();
    double F_integral[NDIM] = {0.0, 0.0, 0.0};  VectorValue<double> F;  libMesh::Point X;  boost::multi_array<double, 2> X_node;
    for (auto el_it = mesh.active_local_elements_begin(); el_it != mesh.active_local_elements_end(); ++el_it)
    {
        Elem* const elem = *el_it;  fe->reinit(elem);  for (unsigned int d = 0; d < NDIM; ++d) dof_map.dof_indices(elem, dof_indices
  [d], d);
        get_values_for_interpolation(X_node, *X_ghost_vec, dof_indices);
        for (unsigned int qp = 0; qp < qrule->n_points(); ++qp)
        {
            interpolate(X, qp, X_node, phi);                    // X at the quadrature point; q_point[qp] is s there
            target_force_function(F, FF_unused, X, q_point[qp], elem, var_data, grad_var_data, loop_time, nullptr);
            for (unsigned int d = 0; d < NDIM; ++d) F_integral[d] += F(d) * JxW[qp];
        }
    }
    IBTK_MPI::sumReduction(F_integral, NDIM);
    const double J_min = IBTK_MPI::minReduction(J_minimum);  const double J_max = IBTK_MPI::maxReduction(J_maximum);  // then reset
    pout << "tether force =";  for (unsigned int d = 0; d < NDIM; ++d) pout << " " << F_integral[d];  pout << "   J from " << J_min
  << " to " << J_max << "\n";
    if (IBTK_MPI::getRank() == 0) { /* append time, Fx, Fy, Fz to data_dump_dirname + "/tether_force.txt" */ }
}
```

THE POST-PROCESSING (2): WHAT THE INTEGRAL DOES

- Chapter 11's recipe in finite elements: a loop over the elements this process owns, a Gauss rule on each (`QGAUSS`, fifth order: more than enough for a force that is linear in $\mathbf{X}$ on a linear element), the force function evaluated at the quadrature points with the interpolated current position $\mathbf{X}$ and the reference position `s` (`q_point`, since `fe` was built on the reference mesh), weights `JxW` of the reference configuration (the force is per reference volume), then a sum over processes.
- It calls `target_force_function` itself rather than reading the `F` system, because the `F` system holds the *total* nodal force (stress plus tether); the horseshoe-crab code (slide 32) integrates the `F` system instead, and so measures the total force the body puts on the fluid.
- The range of J is collected in the stress function on each process and reduced here with a min and a max; a process with no elements of the plate keeps the starting values, which the reductions ignore. After printing, the two are reset for the next interval.
- Only rank 0 writes the file, in append mode, with a header line on the first call; a restarted run continues the file.
- `data_dump_interval` in the `Main` block turns this on (`DATA_INTERVAL`, 100 lines per run); the same hook serves chapter 11's forces and chapter 24's concentration integrals. IBAMR's `ex5` computes its drag the same way; chapter 23 walks through the rest of this file.

THE HORSESHOE CRAB (1): THE “3D IBFE FREE FLOW” EXAMPLE

`3-IBFE-Example-HorseshoeCrab` (course repository, 2018–2020; the Dogwood deck’s “3D IBFE Free Flow”; the zip `2-Example_horseshoe_0deg-13cmps-2018-03`): a horseshoe crab at $0°$ in a stream of 13 cm/s.

- A rigid body in a free stream: no tether band, no flaps; the whole crab is held in place and the stream goes round it. The question is the force on it, the drag coefficient of a shape at a Reynolds number of a few thousand, compared with a flow-tank experiment.
- The mesh is an ExodusII file written by Cubit in December 2017 from a scanned crab: 25864 nodes, 21710 hexahedra, $34 \times 56 \times 12$ mm; “flat” names a flat underside. A body meshed from a scan, not drawn: chapter 18’s third route.
- Units are mm, g and s throughout: lengths in mm, the stream in mm/s, $\rho = 0.001$ g/mm^3 and $\mu = 0.001$ g/(mm s), which is water; a stress in g/(mm s^2) is a pascal.



The crab mesh from above and from the side (VisIt, the `.e` file as shipped).

THE HORSESHOE CRAB (2): THE NUMBERS

tank	$L = 160$: $-80 < x < 80$, $-80 < y < 240$, $-80 < z < 80$ mm; the stream along $+y$; `domain_boxes = [(0,0,0),(N - 1,2*N - 1,N - 1)]`
grid	three levels at ratio 4, $N = 16$: $\mathrm{DX} = 160/256 = 0.625$ mm; about 20 cells across the crab's thickness, 55 across its width
stream	$V\,\dfrac{\tanh((t - t_{1/2})/\tau) + \tanh(t_{1/2}/\tau)}{1 + \tanh(t_{1/2}/\tau)}$ on all six faces, $V = 130$ mm/s, $t_{1/2} = 0.05$ s, $\tau = 0.125$ s: zero at $t = 0$, V by $t \approx 0.4$ s (`VelocityBcCoefs_1`)
Reynolds	$\rho V W/\mu \approx 4400$ with the width $W = 34$ mm; the input's `Re = 400.0` is written but not used (`MU` is set directly)
tether	the whole body: `kappa = kappa_s` for every point (the test `if(s(0)<.2)` is commented out), `KAPPA_S = 1e5`
material	$E = 700$ (stem) and 1400 (leaf) Pa, $\nu = 0.3$, the same formulas as the plate; the leaf values are read and never used
time	`DT = 2e-5`, `END_TIME = 1.0`: 50000 steps; frames every 0.02 s (50), restarts every 0.05 s, data every 0.001 s (1000 lines)
job	`script.sh`: 2 nodes $\times$ 32 tasks $= 64$ cores, 72 hours, on a UNC cluster

- $\mathrm{DT} = 2 \times 10^{-5}$ s with $\mathrm{DX} = 0.625$ mm and $V = 130$ mm/s is a CFL of 0.004; the tether's $\kappa_s = 10^5$ g/(mm^3 s^2) on a body of density 0.001 is what sets it (chapter 19, slide 26).
- The stream along $+y$ and a tank twice as long in y: the crab sits 57 mm from the inflow face and 207 mm from the outflow, with its wake room to develop. The 2017 plate's one unit from the inflow was the lesson (slide 16).

THE HORSESHOE CRAB (3): WHAT THE CODE MEASURES

```
static ofstream forcex_stream, forcey_stream, forcez_stream,
  velx_stream, ...,
               workx_stream, ..., U_L1_norm_stream,
  U_L2_norm_stream, U_max_norm_stream;
...   // in output_data, for each element and quadrature point:
F_integral[d] += F_node[k][d] * phi[k][qp] * JxW[qp];
U_integral[d] += U_node[k][d] * phi[k][qp] * JxW[qp];
W_integral[d] += U_node[k][d]*F_node[k][d] * phi[k][qp] * JxW[qp];

...
forcex_stream << loop_time << " " << F_integral[0] << endl;   //
  F_x.curve
velx_stream   << loop_time << " " << U_integral[0] << endl;   //
  U_x.curve
workx_stream  << loop_time << " " << W_integral[0] << endl;   //
  W_x.curve
```

- Twelve `.curve` files (VisIt's curve format: a time and a value per line, chapter 29): the force on the body integrated from the `F` system (stress plus tether: the total force the body puts on the fluid), the integral of the body's velocity (zero for a body that is held, a check), and $\int U_d F_d$ per component, whose sum over d is the power the body puts into the fluid.
- The L^1, L^2 and max norms of the fluid velocity over the hierarchy (`U_L1.curve`, `U_L2.curve`, `U_max.curve`): a check that the stream has reached its speed and that nothing is blowing up.
- `F_y.curve` is the drag along the stream; `F_x` and `F_z` the side force and the lift. A plot of `F_y` against time with the stream's ramp gives the drag coefficient once the force is steady; chapter 11's recipe.

THE HORSESHOE CRAB (4): WHAT IS WRONG WITH IT, AND WHAT A PORT NEEDS

- `U_integral` is never set to zero: the initialisation loop runs twice for `F_integral` and never for `U_integral`, so the `U_*.curve` files hold whatever was in memory plus the integral. `W_integral` is initialised. The fix is one line.
- Leftovers: `mu_leaf`, `beta_leaf`, `E_LEAF` and `t_set` are read and unused; the material test is commented out; `coordinate_mapping_function` (`X(0) = s(0)`) is defined and never registered; `MESH_FILENAME` defaults to `"Leaf.e"`, a file from another project; `Re` is unused; every input line ends in `//` (the CRLF trick of slide 35).
- Old API: `libMeshEnums::Order`, `SAMRAI_MPI`, `AutoPtr`, `IBAMR_config.h`: chapter 17's list applies before it compiles against 0.16.0 or later; `output_data` then needs `IBTK_MPI` and the current `FEBase::build` as in IBFE/10.
- Cost: 50000 steps of a 256^3-equivalent tank with a 21710-element body on 64 cores for up to 72 hours. It has not been run for this series (IBFE/11 is pending); size it with a short run first, and expect the restart files to matter.
- Angle of attack. "0deg" in the zip's name; other angles come from the coordinate mapping (a rotation of `s`, slide 15) with the tether target rotated the same way, not from new meshes.
- What the port would ship: the mesh, a cleaned `main.cpp` with one `output_data` that writes a single text file (time, three forces, three powers) as IBFE/10 does, an input with the ramp and the tank as above, a Puma job script sized from a timing run, and a README with the first hundred steps' force so that a run can be checked.

LOOKING AT A 3D RUN IN VISIT

1. Open `viz_IB3d/dumps.visit`; **Add** → **Pseudocolor** with the expression `U[0]` (the x-velocity; **Controls** → **Expressions** to define it, or pick the component) and **Draw**: you see the outside of the box.
2. **Operators** → **Slicing** → **Slice**: normal along y $(0, 1, 0)$ through the origin, the x–z plane through the plate's middle. "Project to 2D" makes a flat picture with a 2D view; off, the slice stays in the box for a 3D camera.
3. Open `viz_IB3d/output.ex2`, **Add** → **Mesh**: the plate in its starting position. For the moving plate, the expression `dX = {dX_0, dX_1, dX_2}` (three components now) and **Operators** → **Transforms** → **Displace** with it; a **Pseudocolor** of `dX_0` on the plate shows the bend (slide 23). To cut the plate with the same slice, add the Slice operator to its plot too: its cross-section appears on the 2D picture.
4. The 3D camera: **View** → **3D**, or drag; in a script, `View3DAttributes` with `viewNormal`, `focus`, `viewUp`, `parallelScale` (`wsl/final/visit_ch22.sh`). A **Subset** plot of `levels` shows the grid levels (slide 26).

- `dumps.visit` has `U`, `P`, `Omega` (a vector), `Div_U` and `F`; `output.ex2` has `dX_*`, `X_*`, `F_*` and `U_*` at the nodes. Chapter 27 is 3D visualization (isosurfaces, streamlines, clipping); chapter 28 the finite element data; chapter 30 movies.
- Chapter 21's "Random info" applies: open the fluid first, Draw, then the Exodus file.

WHAT CAN GO WRONG

symptom	cause and fix
`Illegal character token in input` at line 2	the input file has Windows line endings; the course's fix was `//` at the end of every line, the clean one `dos2unix` (chapter 16)
libMesh cannot read the `.msh`, or reads nothing	gmsh 4 wrote MSH 4.1 binary or a file without physical groups; save as Version 2 ASCII with a physical volume (slide 9, chapter 18)
"1D or 2D elements" errors from libMesh	lines and surfaces saved along with the volume; define the physical volume so only it is saved, or `Mesh.SaveAll = 0`
the plate sits at the file's position although a mapping was registered	the mapping was registered after `initializeFEEquationSystems()`, or the tether pulls it back to `s` (slide 15)
the plate near the inflow face does nothing to the flow	it is inside the face's prescribed velocity (the 2017 -7); keep bodies away from prescribed faces (slide 16)
`J from -0.3 to ...`, then NaN	$J < 0$ in a sliver element and $\ln J$ undefined; reduce `DT`, optimise the mesh, lower κ_s
"Max Jacobian 1 / Min Jacobian 1" every step	the course code printed variables nothing updated; IBFE/10 records the real range
the run writes nothing for minutes	3D is slow: 0.6 s per step here; check the log for `At end of timestep` and the first `visit_dump`
`U_x.curve` full of garbage	the crab's uninitialised `U_integral` (slide 33)
`MFAC` changed, nothing happens	it does not apply to a mesh read from a file
the wake hits the outflow face	the outflow velocity is prescribed; lengthen the tank or use a traction condition (chapter 7)

WHAT THE OLDER DECKS HAD THAT IS NOW PART X

2017–2020	now
the tank "$-16 < x < 16$" and the plate shifted by -7	the input was already $-8 < x < 24$; `TRANSLATE_X` as an option, tested (slide 16)
"middle half" with `abs(s(2)) < .25`	`TWO_MATERIALS` with `STEM_HALF_WIDTH` (slide 17)
`vector<NumericVector<double>*>& system_data` (2017)	`var_data` and `grad_var_data` (chapter 19)
`AutoPtr`, `libMeshEnums::Order`, `SAMRAI_MPI`, `IBAMR_config.h`, `abs`	`std::unique_ptr`, `libMesh::Order`, `IBTK_MPI`, no config headers, `std::abs` (chapter 17)
"Void PK1_dil_stress_function" (the 2020 slide 16 typo)	`void`
`data_dump_interval = 1` and an empty `output_data`	100 tether-force lines per run
Makefile with `LIBSNEW`	`CMakeLists.txt`
`//` at the end of every input line	LF line endings
gmsh "Add → Translate → Extrude", "group_name.msh"	Translate → Extrude; the file is named by you (slides 8–9)
the horseshoe crab without slides	slides 30–33; the port is IBFE/11, pending

EXERCISES

1. Continue IBFE/10 to the course's $t = 2$ from its restart files (`END_TIME = 2.0`, `restart_IB3d 5000`). Plot $F_x(t)$ from `tether_force.txt`: when does it stop changing, and what is the steady drag? Compare it with $\rho U^2 D^2 C_D/2$ for a plate at $\mathrm{Re} = 20$.
2. Start the stream impulsively (`gcoef = "U0"`) and compare the first ten force lines with the $\tanh$ start's.
3. `TRANSLATE_X = 4.0`: check in VisIt that the plate sits at $x = 4$ and that `dX_0` reads about 4 there; then `TRANSLATE_X = -7.0` and watch what the inflow face does to the force.
4. `TWO_MATERIALS = TRUE` to $t = 0.5$: how much more do the softer flaps bend? Then fix the E swap and run again.
5. `TETHER_HALF_WIDTH = 0.5`, the "middle half" the 2017 goal said: does the force change, and does the top edge still move more than the bottom one?
6. **Mesh** $\rightarrow$ **Optimize 3D** on `plate2.msh` in gmsh (or extrude in three layers), run to $t = 0.1$, and compare the range of J with the shipped mesh's 0.66–2.51.
7. `Re = 400` (`MU` four times smaller) with `DT = 5e-5`: does a wake leave the fine level's box? Add a threshold or widen the box.
8. Initialise `U_integral` in the crab's `output_data`, remove the leftovers, and estimate from one hundred steps on four cores what the full run would cost.

RULES OF THUMB

- Six faces, the stream on all of them, nothing periodic; the body several kernel widths from any prescribed face, and its wake room to recover before the outflow.
- Volume elements about DX across, two or three through a thin body; a physical volume; MSH 2.2 ASCII; look for slivers before you run.
- The coordinate mapping places a body; whatever compares `x` with `s` must apply the same mapping.
- Time a short run first: 0.6 s per step here on four cores; the course's 2 time units are an afternoon.
- Minus the tether force is the force of the fluid on a held body; an accelerating stream pushes before it flows.
- J far from 1 in a few elements is a mesh defect; $J < 0$ is a run that will end.
- Slice, displace, then camera; the vorticity is a vector in 3D.

SUMMARY

1. IBFE/10 puts a $2 \times 2 \times 0.125$ plate across a stream in a $32 \times 16 \times 16$ tank on three grid levels, holds its middle band with a tether, and lets the flaps bend; the stream is prescribed on all six faces and starts as $\tanh 2t$.
2. The plate's mesh is a gmsh extrusion saved as a physical volume in MSH 2.2; its sliver tetrahedra are where J strays and a few nodes jump.
3. The initial coordinate mapping places the file's mesh in the tank before the first step; `s` stays the file's coordinates, so tethers and prescribed motions must be mapped too. It works in 0.19.0; the 2017 version's trouble was a plate one unit from the inflow face.
4. A 3D run costs 0.6 s per step on four cores and 100 MB per 20 frames; time it, write restarts, and keep the fine level to the body.
5. The tether force, integrated over the plate at Gauss points, is minus the force of the fluid: 0.28 at $t = 0.5$, with the acceleration of the stream in it; 0.16.0 and 0.19.0 agree to six digits.
6. The horseshoe-crab example holds a 25864-node body rigid in a 13 cm/s stream at $\mathrm{Re} \approx 4400$ for 72 hours on 64 cores; its velocity integral is uninitialised and its API is the 2018 one; it is written up here for the first time and ported next.

Chapter 23 walks through a whole IBFE main file, line by line, with IBAMR's `ex7`.

RECORDINGS (LISTED, NOT INCORPORATED)

The 2020 Zoom recording of the IBFE lecture *3D Flow Past Objects* and the 2018 Dogwood session that introduced the flow-tank and free-flow set-ups are listed in the front matter's video list (chapter 0.1). The 2020 deck's render is reproduced on slide 25; the gmsh screenshots on slides 7–9.

REFERENCES AND CREDITS

- Example IBFE/10 of this series (`main.cpp`, `input3d`, `plate2.msh`, `job.slurm`, README with the measured tables); the runs `ft_default`, `ft_translate`, `ft_twomat` with IBAMR 0.19.0 and `ft_default` with 0.16.0; `wsl/final/visit_ch22.sh`. Figures: VisIt 3.5.
- The course material: *7-IBFE 3D Flow Tank* (Feb 2017, edited 2018) and its zip `Example_IBFE_FlowTank3d-mid_plate` (2017); *3-IBFE 3D Flow Past Objects* (2020) and its git history; the Dogwood deck (2018–20, private; numbers only); `examples/legacy/UA-Examples/IBFE Examples/3-IBFE-Example_FlowTank-MidPlate` and `3-IBFE-Example-HorseshoeCrab` (course repository). The gmsh screenshots and the render are the 2020 deck's.
- IBAMR 0.19.0: `src/IB/IBFEMethod.cpp` (`getFromInput`: the `IB_*`, `interp_*` and `spread_*` keys), `examples/IBFE/explicit/ex4` (compressible neo-Hookean, 2D and 3D) and `ex5` (a body held by penalty springs, with drag output). https://github.com/IBAMR/IBAMR
- gmsh: C. Geuzaine and J.-F. Remacle, *Gmsh: a three-dimensional finite element mesh generator with built-in pre- and post-processing facilities*, Int. J. Numer. Meth. Engng 79 (2009) 1309–1331. https://gmsh.info